MILITARY SPECIFICATION

FITTINGS, FLARELESS, FLUID CONNECTION

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 SCOPE: Fittings shall be furnished in the types and styles designated by the applicable Military Standard Detail drawings referred to hereinafter.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified:

SPECIFICATIONS:

FEDERAL

QQ-A-266	Aluminum Alloy 14S; Bars, Rods, Shapes, and Wire - Rolled or Drawn
QQ-A-267	Aluminum Alloy 24S; Bars, Rods, and Shapes, Extruded
QQ-A-268	Aluminum Alloy 24S; Bars, Rods, and Wire, Rolled or Drawn
QQ-A-351	Aluminum-Alloy 75S; Bars, Rods, and Wire (Rolled or Drawn)
QQ-A-355	Aluminum-Alloy (AL-24) (Aluminum-Copper-Magnesium (1.5%)-Manganese; Plates, Sheets, and Strips
QQ-A-367	Aluminum-Alloy; Forgings, Heat-Treated
QQ-M-151	Metals; General Specification for Inspection of
QQ-P-416	Plating; Cadmium (Electrodeposited)
QQ-S-624	Steel; Alloy; Bars (General Purpose)
QQ-S-633	Steel, Carbon; Bars (General Purpose)
WW-T-787	Tubing Aluminum-Alloy 52S, Round Seamless, Drawn

MILITARY

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-F-5506	Fittings: Method of Testing
MIL-H-6088	Heat Treatment of Aluminum Alloys; Process for (For Aircraft Applications)
MIL-H-6875	Heat Treatment of Steels (Aircraft); Process for
MIL-O-5606	Oil; Hydraulic, Aircraft, Petroleum Base
MIL-S-5626	Steel; Chrome-Molybdenum (4140) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-6050	Steel; Chrome-Nickel-Milybdenum (8630) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-6049	Steel; Chrome-Nickel-Milybdenum (8740) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-6758	Steel; Chrome-Molybdenum (4130) Bar, Rod, and Forging Stock (For Aircraft Applications)
MIL-S-7742	Screw Threads, Standard Aeronautical
MIL-T-6845	Tubing; Steel, Corrosion-Resisting, (18-8) Aircraft, Hydraulic System
MIL-T-8504(ASG)	Tubing, Steel, Corrosion-Resistant (18-8) Annealed, Aircraft Hydraulic System
MIL-T-7081	Tubing, Aluminum Alloy (61S), Seamless, Round, Hydraulic System

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JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes; Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-Board and W-Board) Exterior and Interior
JAN-P-120	Packaging and Packing for Overseas Shipment - Cartons; Folding, Paperboard
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials; Waterproof, Flexible
JAN-P-133	Packaging and Packing for Overseas Shipment - Boxes; Set-Up, Paperboard
JAN-P-139	Packaging and Packing for Overseas Shipment - Plywood; Container Grade

AIR FORCE-NAVY AERONAUTICAL

AN-P-13	Preservation and Packaging; Parts and Equipment (General Specification for)

2.2 OTHER PUBLICATIONS:

Bureau of Supplies and Accounts

Navy Shipment Marking Handbook

2.3 DRAWINGS:

Air Force-Navy Aeronautical Standard Drawings

MS21900	Adapter - Flareless Tube to AN Flared Tube
MS21901	Adapter - Flareless Tube for 3/8 Bulkhead and Universal To AN Flared Tube
MS21902	Union - Flareless Tube
MS21903	Union - Flareless Tube 3/8 Bulkhead and Universal
MS21904	Elbow - Flareless Tube 90°
MS21905	Tee - Flareless Tube
MS21906	Cross - Flareless Tube
MS21907	Elbow - Flareless Tube and Universal 45°
MS21908	Elbow - Bulkhead Universal 90° - Flareless Tube
MS21909	Tee - Bulkhead and Universal, Flareless Tube
MS21910	Tee - Flareless Tube - Internal Thread on Side
MS21911	Tee - Flareless Tube, Internal Thread on Run
MS21912	Tube - Flareless Tube With Bulkhead on Run
MS21913	Plug - Flareless Tube
MS21914	Cap - Pressure Seal Flareless Tube Fitting
MS21915	Bushing - Screw Thread Expander, Flareless Tube Connection
MS21916	Reducer - External Thread, Flareless Tube
MS21917	Nut - Sleeve Coupling Flareless
MS21918	Sleeve - Coupling, Flareless
MS33514	Fitting End - Standard Dimensions for Flareless Tube Connection and Gasket Seal
MS33515	Fitting End - Standard Dimensions for Bulkhead Flareless Tube Connections

(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

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3. REQUIREMENTS

3.1 Material. - Fittings shall be fabricated of materials listed in table I as specified on the applicable Military Standard drawing.

3.1.1 Steel. - All steel shall be made by the open hearth or electric furnace process.

3.1.2 Heat Treatment. -

3.1.2.1 Aluminum Alloy. - Aluminum alloy materials shall be heat treated in accordance with Specification MIL-H-6088.

3.1.2.2 When required, steels shall be heat treated in accordance with Specification MIL-H-6875.

3.1.2.3 MS21918, Type A and B sleeves shall be heat treated by the cyanide or similar process to provide a case depth of sufficient thickness to assure adequate grip of the tubing when tested in accordance with paragraph 4.5 to paragraph 4.5.4.3 inclusive of this specification.

3.1.3 Physical Properties. - Unless otherwise specified on the applicable MS detail drawings the hardness of the finished steel parts shall be Rockwell C10-31. For steel fittings falling within the hardness range of Rockwell C10-20, hardness tests shall be made using the Rockwell B scale, hardness values to fall in the range of Rockwell B92-99.

TABLE I
Materials

Material		Specification No.	
Aluminum Alloy	Bars, Shapes, Sheet, Strip and Plate	QQ-A-266	Cond T6
		QQ-A-267	Cond T6
		QQ-A-355	Cond T6
	Forgings	QQ-A-367 Class 5	Cond T6
Steel	Bars, Rods, and Forgings	MIL-S-6758	
		AN-S-9	
		MIL-S-6050	
		QQ-S-633	(FS1117
			(FS118
			(FS1137
			(FS1335
			(FS1141
			(C1212
		QQ-S-624	
		MIL-S-6049	

3.1.4 Alternate Material. - Materials not listed in table I and use of which has not been specifically prohibited may be used only upon specific approval of the Bureau of Aeronautics. In order to obtain this approval it will be necessary to conduct complete Performance tests for alternate materials and submit a test report as specified herein. In no case shall the alternate material be used in production fittings prior to receipt of specific approval from the Services.

3.1.5 Except when specifically allowed on the MS Standard detail drawing, castings of any type shall not be used.

3.1.6 Fittings made by welding or brazing together two or more separate pieces of metal may be used on service aircraft only when the design method of fabrication by each manufacturer offering such fittings has been submitted to and approved by the Bureau of Aeronautics.

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3.2 Design and Dimensions. - The design and dimensions of Military Standard fittings shall be in accordance with the applicable Military Standard detail drawings.

3.2.1 Shape and Form Restrictions. - Abrupt reductions of section shall be avoided. Small external sections adjoining relatively heavy body sections shall be shaded into the heavier sections by means of ample fillets. Sharp corners or inadequate fillets, excessive undercuts or excessive grooves at the junction of such small sections with large sections of fittings shall be considered cause for rejection. Unless otherwise specified, drawing tolerances shall apply to fillet and corner radii.

3.2.2 Passages. -

3.2.2.1 Drill Offset. - On straight fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed 0.015 inch. It shall be possible to pass through the fluid passage a ball whose minimum diameter is 0.020 less than the minimum diameter specified for the passage. This does not mean that the drilled passage may be smaller than that required by the detail drawings.

3.2.2.2 On angle fittings, the cross-sectional area at the junction of the fluid passages shall be not smaller than the cross-sectional area of the smaller passage.

3.2.2.3 Wall Thickness. - Except as otherwise specified on the applicable MS drawing, the wall thickness at any point on the fitting shall be not less than the thickness established by the dimensions and tolerances for the inside and outside diameters and eccentricities specified in the MS detail drawing.

3.2.2.4 Threads. - Except as otherwise specified, all straight threads shall conform to Specification MIL-S-7742.

3.3 Finish. - All fittings shall be finished as noted in table II in accordance with the applicable finish specification, and as specified in paragraphs 3.3.1 thru 3.3.1.7, inclusive.

TABLE II
Finish

Fitting Material	Type of Finish	Finish Specification
Aluminum-Alloy	Anodic Treatment	MIL-A-8625, Type II
Steel	Cadmium Plate	QQ-P-416, Type II, Class C

3.3.1 Anodic Treatment of Aluminum Alloy Flareless Fittings. - Supplementing Specification MIL-A-8625, aluminum alloy fittings incorporating MS33514 and MS33515 ends shall be subject to the following requirements:

3.3.1.1 Etching. - Aluminum alloy fittings shall not be etched after machining.

3.3.1.2 Color. - Aluminum alloy fittings shall not be artificailly colored by dyeing.

3.3.1.3 Sulphuric Acid Concentration. - Shall be 15% to 18% by volume.

3.3.1.4 Current Density. - Anodic current density shall be 12 to 16 amperes per square foot.

3.3.1.5 Time. - Aluminum alloy fittings shall be anodized for 30 minutes.

3.3.1.6 Continuity of Coating. - The anodic coating of internal surfaces, ends, threads and thread reliefs shall be clean, unbroken and free from rack contact marks.

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3.3.1.7 Sealing. - Fittings shall be sealed in 1% to 5% potassium dichromate solution by weight, for 10 minutes at 210°F.

3.4 Identification of Product. - Unless otherwise specified all fittings shall be permanently marked with the letters MS and the manufacturer's name or trademark. If impression stamping is not used the method of marking shall be subject to the approval of the Bureau of Aeronautics. The letters MS shall be separate and distinctive from the manufacturer's name or trademark. This marking shall be applied in a location not detrimental to the fitting and shall not be detrimental to the corrosion protection of the fitting. In addition, MS Standard fittings shall be identified by the following coloring:

Aluminum-Alloy	- No dye
Steel Fittings	- Natural Cadmium Plate
Steel Ferrules	- Natural Cadmium Plate

3.5 Weight. - The maximum weight of fittings shall not exceed the values on the applicable MS drawings.

3.6 Workmanship. - Fittings shall be free from cracks, laps, seams, burrs, longitudinal and spiral tool marks, or any other defects which may detrimentally affect their suitability for the service intended. Contractor's production and inspection processes shall be established so the end results will produce a completed article in strict compliance with specification and contract requirements.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests. - The inspection and testing of flareless fittings shall be classified as follows:

(a) Qualification tests; Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on fittings manufactured and submitted for acceptance under contract.

4.2 Qualification Tests. -

4.2.1 Sampling Instructions. - Qualification tests samples shall be submitted to the Service designated in paragraph 6.3, plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification Test
FITTING FLARELESS, FLUID CONNECTION
MS Part No.
Name of Manufacturer
Manufacturer's Part No.
Submitted by (name) (date) for Qualification test in accordance with the requirements of Specification MIL-F-18280 under authorization (reference authorizing letter).

4.2.2 Tests. - Qualification tests of fittings shall consist of all the tests of this specification. The Qualification test may, at the option of the qualifying agency, be supplemented with tests under actual service conditions.

4.3 Inspection Tests. - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. Inspection test samples shall be in addition to the quantity specified on the contract or purchase order and shall be furnished without additional cost to the Government. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not

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having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Material Certification. - Certificates showing conformance with the applicable material specification shall be available to the Inspector for each lot of fittings.

4.3.2 Examination of Product. - All fittings shall be examined to determine conformance with this specification with respect to the requirements of workmanship, marking, and finish.

4.3.3 Sampling Tests. - A maximum of 1 percent, but not less than 10 of the fittings from each lot on a contract, shall be selected at random for the inspection tests specified herein. A lot shall consist of all fittings of one type and size made from the same batch of material and submitted for inspection at the same time and place. When the sample fittings are not damaged by the tests, they may be included in the contract or order. Samples shall be subjected to the tests described herein for Examination of Product, and conformance to dimensions and tolerances specified on the applicable MS Standards, and on the manufacturer's detail drawings of the design upon which Qualification test approval was based.

4.3.4 The Government Inspector may require magnetic inspection or other Inspection tests whenever Government Service experience warrents.

4.3.5 Rejection and Retest. - If 10 percent of the sample fittings fail to conform to any of the requirements of this specification the whole lot shall be tested to those requirements in which the 10 percent of sample fittings failed. Those fittings subjected to individual tests and failing to meet the requirements shall be rejected. Fittings which have been rejected may be reworked to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Service.

4.4 General Testing Practice for Qualification Tests. -

4.4.1 Thread Lubricant. - No thread lubricant shall be used on straight thread fittings other than hydraulic fluid conforming to Specification MIL-O-5606 unless specifically approved by the Bureau of Aeronautics.

4.4.2 Tightening Torques. - Normal tightening torques of the proposed fittings shall conform to the values established by the manufacturer. Minimum and maximum values shall be established for each size and general type of tube material. Torque values for "over-tightened" assembly conditions for all sizes shall be 2-1/2 times the minimum value listed.

4.4.3 Tube Material and Working Pressures. - Fittings shall be tested with the tube materials specified in table III. The operating pressures listed in table III shall be considered normal working pressures for high pressure fittings.

4.4.4 Test Fluid. - Fluid conforming to Specification MIL-O-5606 shall be used in all tests.

4.4.5 Number of Samples Tested. - A minimum of six separate fitting assemblies or three sets of tube assemblies with fittings at each end, shall be used for each test.

4.5 Qualification Testing of Fittings. - Qualifications Tests shall include the following

(a) Repeated Assembly Tests
(b) Impulse Tests
(c) Pressure Tests
(d) Vibration Tests

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TABLE III
Tube Material and Pressures
1/

Tube Materials	Aluminum Alloy 61S-T6			Stainless Steel 18-8		
Specification	MIL-T-7081			MIL-T-6845		MIL-T-6846
Condition	Condition T			1/4 Hard		Annealed
Tube OD Nominal	Size	Wall	Work Press	Wall	Work Press	Wall	Work Press
1/8	2	.020	3000	.016	3000	.020	3000
3/16	3	.028	3000	.018	3000	.020	3000
1/4	4	.035	3000	.020	3000	.028	3000
5/16	5	.049	3000	.022	3000	.028	3000
3/8	6	.049	3000	.025	3000	.035	3000
1/2	8	.065	3000	.028	3000	.049	3000
5/8	10	.083	3000	.035	3000	.058	3000
3/4	12	.095	3000	.042	3000	.065	3000
1	16	.065	1500	.065	3000	.083	3000
1-1/4	20	.083	1500	.065	3000	.065	1500
1-1/2	24	.095	1500	.095	3000	.065	1500
1-3/4	28	.120	1500	.095	3000	.095	1500
2	32	.156	1500	.095	3000	.095	1500

4.5.1 Repeated Assembly Tests.-

4.5.1.1 Minimum Torque. - The fitting assemblies shall be tightened fifteen successive times using the minimum tightening torque recommended for the fittings being tested. Each tightening operation shall include a complete removal of the tube from the fitting body. After each third tightening operation, the fitting assemblies shall be subjected to fluid pressure of 2 times the working pressure (held for 5 minutes). There shall be no leakage of the fitting assemblies when subjected to these pressures. At no time during the fifteen tightening operations shall the assemblies be difficult to assemble or disassemble. After the fifteenth tightening operation, there shall be no leakage or blowoff of the fitting assemblies up to the burst of the tubing or to 4 times the working pressure, (held for 5 minutes), whichever is the lower pressure.

4.5.1.2 Maximum Torque. - The tests specified under "Minimum Torque" shall be repeated with the same results except that the fitting assemblies shall be tightened to the maximum wrench torque recommended for the proposed fittings.

4.5.1.3 Overtightened Torque. - The test specified under "Minimum Torque" shall be repeated with the exception that the fitting assemblies shall be tightened to the Over-tightened Torque as specified in the paragraph entitled "Tightening Torques" under "General Testing Practice." The assemblies shall not leak or blowoff when the test pressures are applied, but moderate difficulty in assembly and disassembly shall not be cause for rejection.

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4.5.1.4 Torque on Tubing During Assembly. - The tendency for the tube to turn during assembly shall be recorded and shall not exceed the values given in table IV.

TABLE IV
Turning Torque on Tube
inch-Pounds (MAX)

Tube Size	2	3	4	5	6	8	10	12	16	20	24	28	32
Torque	5	5	5	5	110	15	20	20	25	25	25	25	25

4.5.1.5 Minimum Opening. - At no time shall any of the assemblies tested under "Minimum and Maximum Torque" present an internal diameter opening through either fitting or tube less than those specified in table V.

TABLE V
Minimum Opening

Tube Size	Minimum Opening
2	.049
3	.112
4	.159
5	.221
6	.284
8	.378
10	.471
12	.596
16	.820
20	1.065
24	1.299
28	1.534
32	1.768

4.5.2 Impulse Test. - All impulse tests are conducted at a rate of 35 ±5 cycles per minute. Each impulse pressure shall constitute a rise from 0 psi to working pressure and drop to zero. When hydraulic fluid is used, a peak surge pressure of 1.43 to 1.57 times the working pressure, as shown by an oscilliscope, shall be obtained. Assemblies at each of the torque conditions noted below shall be tested at room temperature for 100,000 impulse cycles, and at -54°C (-65°F) for 100,000 cycles. The assemblies shall be subjected to 2 to 10 pounds psi static pressure for 24 hours at -54°C (-65°F) prior to the Low Temperature Impulse test; this time period shall be increased to 72 hours if the fitting assemblies contain non-metallic parts. During this cold soak period, there shall be no leakage. Before and after the impulse tests at both room temperature and -54°C (-65°F) the fitting assemblies shall be tested at proof pressure for 5 minutes without any fluid leakage or fitting failure.

4.5.2.1 At Minimum Torque. - Assemblies shall be tightened to applicable minimum tightening torques and subjected to the Impulse test. There shall be no leakage or failure of any of the test assemblies.

4.5.2.2 At Overtightened Torque. - Assemblies shall be tightened to the overtightened wrench torque as specified in the paragraph entitled "Tightening Torques" and subjected to the Impulse test. There shall be no leakage or failure of any of the test assemblies.

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4.5.3 Pressure Test. - Each body shape, size, and material shall be subjected to 4 times the operating pressure with no sign of leakage, distortion, or burst.

4.5.4 Vibration Test. - The vibration test shall consist of mounting a fitting and tube assembly, as noted below, in a vibration machine as specified below, and stressing the tube, in a vibration movement, with respect to the fixed end of the fitting. The ability of the test fitting to withstand vibration will be compared to that of an AN Standard fitting.

4.5.4.1 Fitting and Tube Assemblies Used. - A length of tubing, of a length shown in table VI, straight to within ±.002 inches, shall be assembled to the fitting at an end of the tubing. For installation of the AN fitting, an AN815 type fitting with AN818 Nut and AN819 Sleeve shall be assembled on tubing flared in accordance with AND10061 or AND10079 using the minimum wrench torques specified in AND10064. Flareless tube fittings being tested shall be assembled with the minimum wrench torque recommended for the fitting. All fittings shall be mounted into a boss in accordance with AND10050 on the vibration machine.

4.5.4.2 Vibration Machine. - The vibration machine shall be designed so that the tube assemblies with fittings at each end can be mounted as follows:

(a) One end shall be rigidly mounted.
(b) The other end shall be mounted so that it shall be free to slide in one plane.

The machine shall have provisions to displace one of the ends from the axial line through fitting and tube, thus stressing the specimen at a speed of 1,750 ±50 cpm. It shall be possible to apply impulse pressure from 0 psi up to the operating pressure at a rate of 35 ±5 cycles per minute, to the fitting and tube assembly during vibration.

4.5.4.3 Conducting Vibration Tests. - The fitting and tube assembly shall be mounted in the vibration machine and the total axial displacement and length of specimen, shown in figure I of Specification MIL-F-5506, shall be such as to stress tubing to approximately 1/3 of its ultimate tensile strength, with the effective tube length calculated as a cantilever beam of constant moment of inertia.

The specimen shall be subjected to vibration, while impulse pressures are being simultaneously applied until failure occurs or until completion of 10,000,000 vibration cycles. The average number of cycles before failure for each size of flareless tube fitting shall be the same, or more than, the average number of cycles withstood by the corresponding size of AN tube fitting. Not more than one specimen of each size of a flareless tube fitting shall have a lower life than the lowest vibration life of the corresponding AN fitting. The results of the tests of the AN fittings and the proposed tube fittings shall be listed in a table such as table VII.

TABLE VI
Displacement and Length of Tubing

	Size	3	4	5	6	8	10	12
5280	Length "L"							
	Displacement							
18-8 Annealed	Length "L"							
	Displacement							
61S-T6	Length "L"							
	Displacement							
18-8 1/4 hard	Length "L"							
	Displacement							

NOT AVAILABLE TO BE CALCULATED AND ENTERED HEREIN

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TABLE VII
Vibration Performance

		Flareless Tube Fittings							AN Tube Fittings
	Size	3	4	5	6	8	10	12	
52SO	Min. Cycles								
	AV. Cycles								
18-8 Annealed	Min. Cycles								
	Av. Cycles								
18-8 1/4 hard	Min. Cycles								
	Av. Cycles								
61S-T6	Min. Cycles								
	Av. Cycles								

4.5.5 Additional Tests. - Additional tests may be performed by the Services if considered necessary.

5. PREPARATION FOR DELIVERY

5.1 Application. - The packaging, packing, and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 Preservation and Packaging. - Fittings in unit quantities specified by the Procuring Service shall be protected from corrosion in accordance with method IA of Specification AN-P-13 except that no preservative compound shall be applied to the fitting. Containers conforming to Specification JAN-P-120, JAN-P-133 or JAN-P-108 shall be used in the preservation process.

5.3 Packing. - Unless otherwise specified, all items shall receive domestic packing. Shipping containers, shall contain the same number of unit packages, shall be uniform in size, and snugly packed. The gross weight of the fully packed container shall not exceed approximately 200 pounds.

5.3.1 Domestic Packing. - Unless otherwise specified, interior packages shall be overpacked in substantial commercial exterior containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. The use of corrugated or solid fiberboard having a minimum Mullen test of less than 275 pounds is prohibited. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3.2 Overseas Packing. - Unless otherwise specified, for overseas shipment, the interior containers shall be packed in boxes, in accordance with Specification JAN-P-105 or JAN-P-106. Plywood, if used, shall be in accordance with Specification JAN-P-139 type B, condition I. A sealed case liner conforming to Specification JAN-P-125 is required.

5.4 Marking and Labeling. - Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

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FITTINGS (see applicable AS drawing for title)
AS Part No.
Specification MIL-F-18280(Aer)
Stock No.
Manufacturer's Drawing No.
Name of Manufacturer
Name of Contractor (if different from manufacturer)
Contract or Order No.
Quantity

5.4.1 Shipping Containers. - Each shipping container shall be marked in accordance with the requirements specified in the Navy Shipment Marking Handbook for the Navy.

6. NOTES

6.1 Intended Use. - Flareless fittings covered by this specification are intended for use in aircraft.

6.2 Ordering Data. - Requisitions, contracts, and orders should state whether overseas packing is desired. (See Section 5.)

6.2.1 Qualified Products Lists. - Fittings considered acceptable under this specification are listed in the latest revision of QPL-18280(Aer).

6.3 Provisions for Qualification Tests. - The right is reserved to reject any bids on fittings which have not been subjected to the required qualification tests and found satisfactory. The attention of manufacturers is called to this provision, and the need to request authorization for tests of the fittings which they propose to offer to the Navy under this specification. Requests for authorization of tests, together with two certified test reports giving a detailed description of the tests conducted and a discussion of the results obtained showing conformance with all requirements of this specification, and the manufacturer's assembly and detail drawings, shall be submitted to the Bureau of Aeronautics, Navy Department, Washington 25, D. C. It is to be understood that the sample or samples will be furnished at no cost to the Government and that the manufacturer shall pay all transportation charges to and from the point where tests are made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the procuring agency considers sufficient to warrant additional tests.

6.3.1 It is to be understood that fittings supplied under contract shall be identical in every respect to the sample tested and found satisfactory and to the detailed drawings thereof, except for changes as previously approved by the procuring agency. Any unapproved changes from the qualification sample or from the drawings shall constitute cause for rejection.

6.3.2 After successfully passing all qualification tests, the manufacturer's name will be placed on QPL 18280(Aer).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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